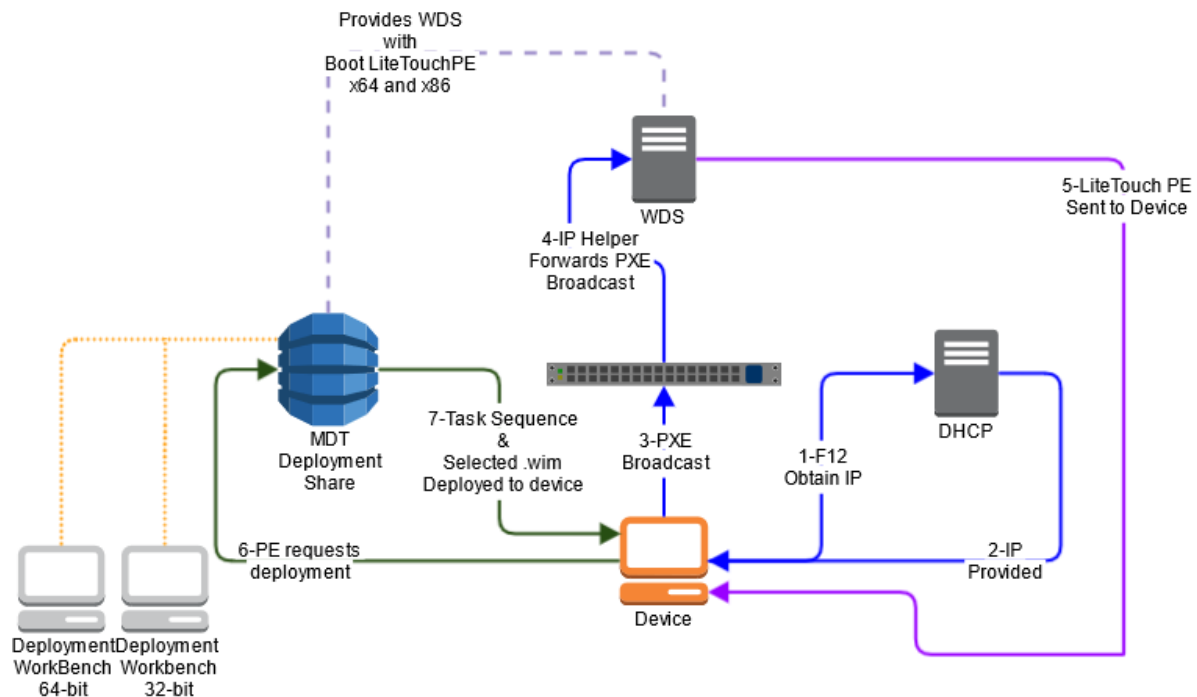


Create and manage deployment image using MDT-11



1. Introduction

Microsoft Deployment Toolkit (MDT) is a free tool from Microsoft used to automate deployment of Windows operating systems. MDT helps administrators create standardized images and deploy them across multiple systems efficiently.

2. Requirements

- Windows Server 2019/2022 with Active Directory
- Microsoft Deployment Toolkit installed
- Windows ADK (Assessment and Deployment Kit) installed
- A domain-joined machine to act as MDT Server
- Windows ISO (for the OS image to deploy)

3. Install MDT and ADK

1. Download and install **Windows ADK** with Deployment Tools.
2. Download and install **Windows PE add-on for ADK**.

3. Install **Microsoft Deployment Toolkit**.
4. After installation, open **Deployment Workbench**.

4. Create Deployment Share

1. In Deployment Workbench:
 - Right-click **Deployment Shares** > **New Deployment Share**.
2. Set path, for example:
3. D:\DeploymentShare
4. Provide a share name:
5. DeploymentShare\$
6. Configure options:
 - Ask for a product key: No
 - Ask for admin password: Yes
 - Ask for computer name: Yes
 - Ask for local admin password: Yes
7. Finish setup.

5. Import Operating System

1. In Deployment Workbench:
 - Right-click **Operating Systems** > **Import Operating System**.
2. Choose source:
 - Full set of source files (Windows ISO extracted).
 - Example: D:\Win10ISO.
3. Destination directory: Win10Ent.

6. Create Task Sequence

1. Right-click **Task Sequences** > **New Task Sequence**.
2. Enter details:

- Task Sequence ID: WIN10TS
 - Task Sequence Name: Windows 10 Enterprise
3. Select **Standard Client Task Sequence**.
 4. Select imported OS.
 5. Do not specify product key (unless required).
 6. Provide local admin password.
 7. Finish wizard.

7. Update Deployment Share

1. Right-click Deployment Share > **Update Deployment Share**.
2. This generates boot images (LiteTouchPE_x64.wim and ISO).
 - Location: D:\DeploymentShare\Boot.
3. These boot images are used to start client machines.

8. Deploy Image to Client

1. Boot target machine using PXE or LiteTouch ISO.
2. Select Task Sequence (e.g., Windows 10 Enterprise).
3. Deployment starts:
 - Partition disk
 - Install Windows
 - Install drivers and applications
 - Apply settings
4. System restarts and joins the domain if configured.

9. Manage Deployment Images

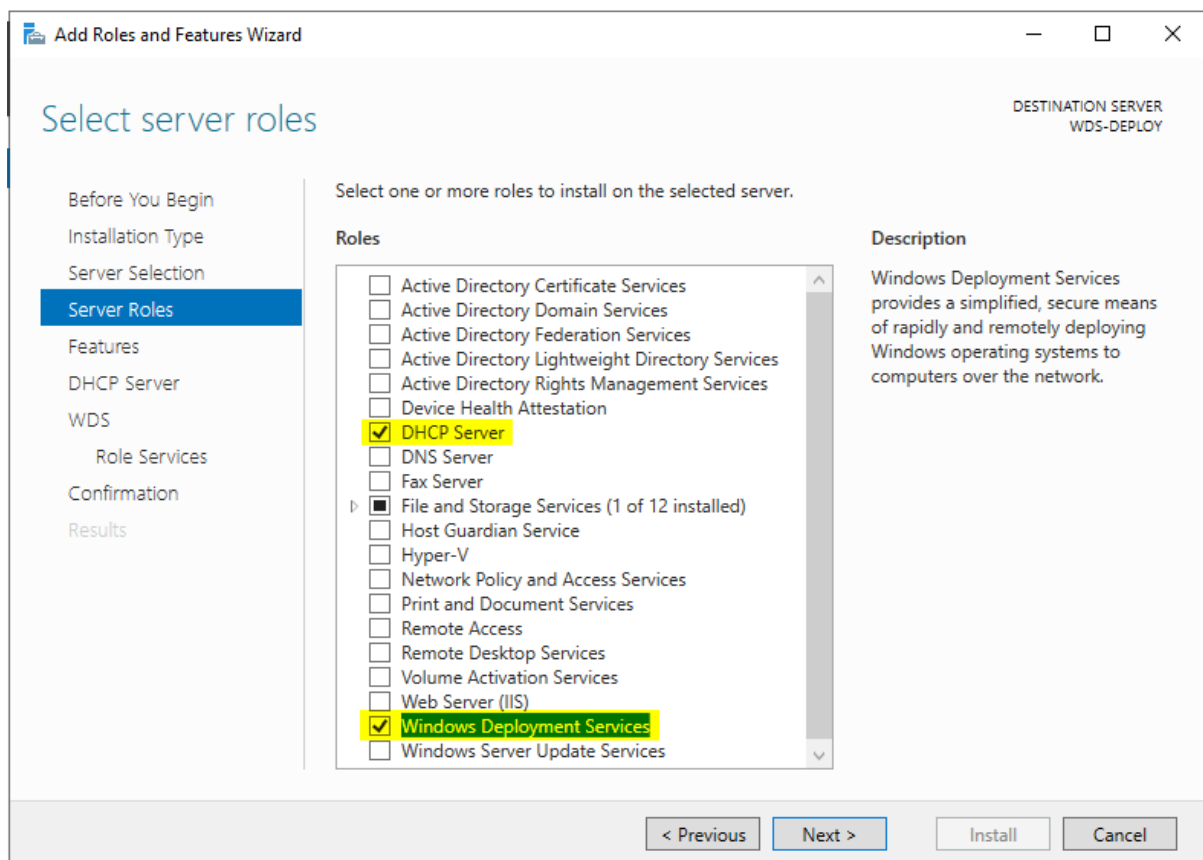
- **Update Deployment Share** after adding drivers, apps, or OS.
- **Capture a reference image:**
 1. Create a Task Sequence for capture.

2. Deploy to a VM.
 3. Customize OS (apps, updates).
 4. Run capture process.
 5. Store .wim in Deployment Share.
- Use captured image for faster standardized deployments.

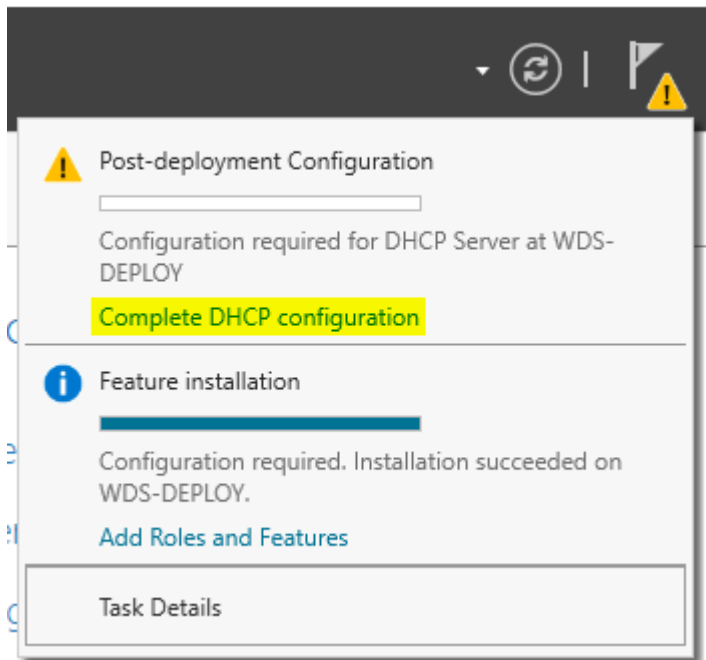
WDS Deployment Using VMware

1. Install the WDS Role

- Open **Server Manager** → Manage → Add Roles and Features.
- Select **Role-based or feature-based installation**, choose your server.
- Under **Server Roles**, check **Windows Deployment Services** (include Deployment Server and Transport Server).
- Click Install.



2. Configure WDS



- Open **Windows Deployment Services** from Server Manager's Tools menu.
- Right-click your server → **Configure Server**.
- Choose **AD-integrated** (recommended for domain environments like wipro.com) or Standalone if not domain joined.
- Set the **Remote Installation Folder** (e.g., D:\RemoteInstall).
- Configure **PXE response behavior** (e.g., respond to all clients automatically).

3. Add Boot and Install Images

- Mount or extract your Windows Server or Windows 10 ISO.
- In the WDS console:
 - Right-click **Boot Images** → **Add Boot Image**, point to boot.wim.
 - Right-click **Install Images** → **Add Install Image**, point to install.wim (create an Image Group if needed).

4. Configure DHCP Settings

- Ensure a working DHCP server exists on your network.
- If running DHCP and WDS on the same server, configure DHCP options 66 (Boot Server Host Name) and 67 (Bootfile Name) to point to the WDS server. **5. PXE Boot and Deploy**
- Boot a VM via PXE (enable network boot in BIOS).

- Press F12 if required, or let it auto-boot per WDS settings.
- The VM should receive an IP, load boot.wim via network, then present available install images.
- Select your Windows image and install; the deployment will proceed automatically.